5. Scenario: You are a data analyst working for a car manufacturing company. As part of your analysis, you have a dataset containing information about the fuel efficiency of different car models. The dataset is stored in a NumPy array named fuel_efficiency, where each element represents the fuel efficiency (in miles per gallon) of a specific car model. Your task is to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models.

Code;

```
import numpy as np

# Fuel efficiency of car models in MPG

fuel_efficiency = np.array([25, 30, 35, 40, 45])

# Calculate average fuel efficiency
average = np.mean(fuel_efficiency)

# Calculate percentage improvement
improvement = ((fuel_efficiency[1] - fuel_efficiency[0])
        / fuel_efficiency[0]) * 100

# Display results

print("Fuel Efficiency:", fuel_efficiency)

print("Average Fuel Efficiency:", average, "MPG")

print("Percentage Improvement:", improvement, "%")
```

INPUT;

File Edit Format Run Options Window Help

```
import numpy as np

# Fuel efficiency of car models in MPG
fuel_efficiency = np.array([25, 30, 35, 40, 45])

# Calculate average fuel efficiency
average = np.mean(fuel_efficiency)

# Calculate percentage improvement
improvement = ((fuel_efficiency[1] - fuel_efficiency[0])
               / fuel_efficiency[0]) * 100

# Display results
print("Fuel Efficiency:", fuel_efficiency)
print("Average Fuel Efficiency:", average, "MPG")
print("Percentage Improvement:", improvement, "%")
```

OUTPUT;

IDLE Shell 3.13.14

File Edit Shell Debug Options Window Help

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>
========== RESTART: C:/Users/Jashw/exxxxxxxxxxpppp4 fds.py ==========
Quarterly Sales: [10000 12000 15000 20000]
Total Yearly Sales: 57000
Percentage Increase: 100.0 %
>>
========== RESTART: C:/Users/Jashw/exxxxxxxppp5 fds.py ==========
Fuel Efficiency: [25 30 35 40 45]
Average Fuel Efficiency: 35.0 MPG
Percentage Improvement: 20.0 %
>>
```